Insights into the neurobiology of Behavioral Inhibition from Nonhuman Primate models

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Introduction

Imagine you're in a room and suddenly, a tiger enters, however, it is unaware of your presence. What would you do? If your instinct is to stay quiet and avoid drawing attention to yourself, you're likely increasing your chances of surviving the encounter. Although the scenario of a tiger strolling into a room is unlikely for most of us, we intuitively understand that inhibiting our behavior in this context can increase our chances of survival. Throughout evolution, animals that adaptively respond to their environment are more likely to survive (Fox, 2018). Now, let's shift the scenario slightly. Imagine that you're in a room filled with unfamiliar people. Although experiencing some amount of anxiety can be adaptive, extreme levels of anxiety and behavioral inhibition in this context can be maladaptive, leading to psychological distress and rob people of the opportunity to form new relationships (Clauss & Blackford, 2012; N. A. Fox et al., 2021; Kagan et al., 1984, 1988; Schwartz et al., 2003). In short, although inhibiting behavior can be adaptive in some contexts and even increase the chances of survival, extreme responses in other contexts can be maladaptive.

Some children react with extreme behavioral inhibition (BI), tending to inhibit their behavior when faced with potentially threatening or novel stimuli (Clauss & Blackford, 2012; Kagan et al.,1984, 1988). Extreme BI can be maladaptive, and decades of work has shown that children who show extreme levels of BI are at risk to develop anxiety and depressive disorders (Biederman et al., 2001; Clauss et al., 2014; Clauss & Blackford, 2012; N. A. Fox et al., 2005; Kagan et al., 1988). Childhood BI is stable over development, marked by heightened physiological reactivity, and partially heritable (Clauss et al., 2015; N. A. Fox et al., 2021; Kagan et al., 1987; Schwartz et al., 2003). It is estimated that approximately 15-20% of children display BI (Clauss et al., 2015; Kagan et al., 1988). Understanding the neurobiology of BI provides a unique opportunity to guide the development of early-life behavioral, cognitive, and pharmacological interventions to prevent the onset of anxiety and depressive disorders.

Optimal progress towards elucidating the neurobiology of BI requires generating and testing novel hypotheses that are best suited for studies in animal models. In humans, BI is often accompanied by subjective feelings of anxiety and apprehension. Although subjective feelings themselves are only overtly measurable in humans, these feelings emerge from distributed anatomical and molecularly diverse neural circuits that are associated with BI (Etkin et al., 2011; A. S. Fox & Shackman, 2019; Pine et al., 2021). Because BI can be important for survival, BI can be observed in many different species making it amenable to cross-species study (Cavigelli & McClintock, 2003; Clauss et al., 2015; Kalin & Shelton,

2003; Qi et al., 2010). In this section, we outline how nonhuman primates (NHPs) can be leveraged as a model organism for understanding human BI, and how we can leverage NHP models to perform both discovery-based and mechanistic studies to elucidate the neurobiology of BI and inform our understanding of the risk to develop stress-related psychopathology.

Modeling Behavioral Inhibition in NHPs

The neurobiology of BI in humans includes brain areas that are, in part, evolutionarily-conserved and have been elaborated throughout evolution. The large evolutionary gap between rodents and humans often limits the direct clinical applicability of rodent findings (Azkona & Sanchez-Pernaute, 2022; Belmonte et al., 2015). However, due to their phylogenetic proximity to humans, NHPs are a crucial animal model in neuroscience, helping bridge the gap between rodent and human studies. NHPs and humans share many biological, cognitive, and social similarities, enhancing the translatability of NHP findings to humans (Chang & Platt, 2014; Dunbar & Shultz, 2007; Kalin & Shelton, 2003; Miller et al., 2016).

In general, NHPs can be classified into several simian species, including monkeys and apes, as well as prosimians such as lemurs (Welker, 2017). Monkeys can be further categorized into two groups – Old World (Catarrhini) and New World (Platyrrhini) monkeys. Old World monkeys are native to Africa and Asia and include species such as macaques and baboons. New World monkeys are native to the Americas, and include species like marmosets and titi monkeys. Old World Monkeys diverged most recently from humans approximately 25 million years ago (MYA), whereas New World monkeys diverged approximately 35 MYA. Importantly, both Old- and New-world primate species diverged from humans much more recently than rodents, which diverged from humans ~75-90 MYA (**Fig 1**).

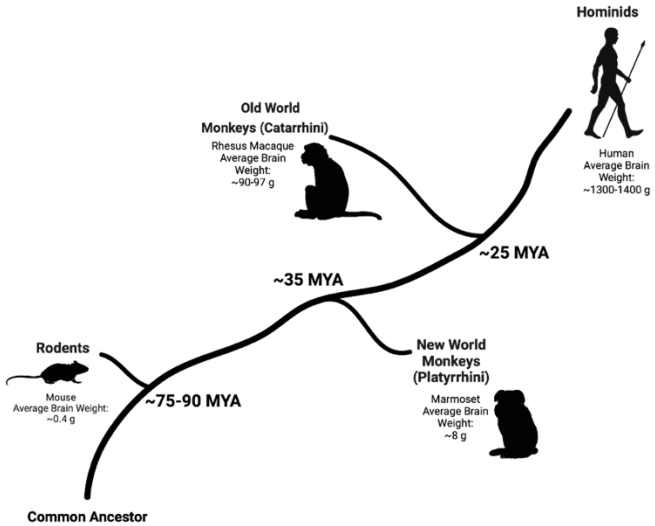

Figure 1: Phylogenetic tree showing evolutionary divergence of humans from rodents and NHP species. Created with BioRender.com

This recent evolutionary divergence between humans and NHPs holds significance for neuroscience, as it is reflected in the composition and function of the molecules, cells, and circuits that give rise to complex behavior in humans. This is most notable in the prefrontal cortex (PFC), as NHPs and humans share highly elaborated PFC regions, which have been implicated in anxiety and emotion (Birn et al., 2014; Fox & Kalin, 2014; Öngür & Price, 2000; Petrides & Pandya,1991; Pine et al., 2021). However, similarities between NHPs and humans extend far beyond the PFC, as many brain regions which are generally considered to be evolutionarily conserved contain primate-specific cell-types, which may contribute to human behavior (Kamath et al., 2022; Krienen et al., 2020; Schmitz et al., 2022). The shared biology between NHPs and humans underlies similarities across developmental (e.g. Paus et al., 2001), social (e.g. Dunbar & Shultz, 2007), affective (e.g. Fox & Kalin, 2014), and cognitive (e.g. Rich & Wallis, 2016) functions -- including BI.

As might be expected from behaviors that increase the likelihood of survival, BI is not a distinctly human phenomenon; heightened reactivity to novel stimuli is widely observed across species (Fox et al., 2008; Qi et al., 2010). Over the years, researchers have developed various models to study the different

facets of BI in animal species (Chun & Capitanio, 2016; Clauss et al., 2015; Kalin & Shelton, 2003). Due to the biological similarities in neurobiology alluded to above, NHPs are particularly well-suited for this purpose, as NHPs brains reflect the integration between rodent-conserved emotion-related pathways with more recently expanded cortical areas, which can work in concert to increase the chances of survival (Fox, 2018). As such, the shared biology between NHPs and humans can be leveraged to provide insight into the complex socio-emotional processes associated with BI. Here, we review rhesus monkey models of BI, which focus on heightened reactivity to novel or potentially threatening stimuli, and operationalized as trait-like tendencies toward increased timidity, freezing, or withdrawal responses. Just as in humans, rhesus monkey BI models require developmentally appropriate methods to quantify variation in BI. Below, we outline two such models of BI, focusing on the peri-adolescent and infant periods.

Peri-adolescent Anxious Temperament (AT)

The Human Intruder Paradigm (HIP) was developed in rhesus macaques by Kalin and Shelton to model earlier studies done in children (Kalin & Shelton, 1989), but has since been adapted by many groups using other NHPs (Pryluk et al., 2020; Rahman et al., 2021). The HIP consists of various phases, each meant to elicit a specific behavioral response from the animals. During the *alone* phase, the animal is separated from conspecifics and placed alone in the testing environment. In this phase, animals display increased activity and cooing behaviors. These are believed to be affiliative behaviors, meant to recruit help from conspecifics in this unfamiliar environment. During the *stare* phase, an experimenter enters the room and makes direct eye contact with the animal, often inducing aggressive behaviors, including “barking” and teeth-bearing facial expressions.

The “no eye contact” (NEC) phase of the human intruder paradigm is specifically designed to elicit behaviors commonly used to assess BI. In this phase, an experimenter enters the room, presents their profile to the test subject while making no eye-contact, and remains motionless. During NEC, animals often freeze, inhibiting their movement and vocalizations. Like the previous example with a tiger, this behavior can function to reduce the chances of predator detection. Freezing behavior during NEC is often accompanied by decreases in coo vocalizations (Fox et al., 2005, 2008), and increased cortisol levels (Kalin et al., 1998), consistent with broad-scale changes in behavior and physiology seen in childhood BI. This composite phenotype (increased freezing, decreased cooing, and increased cortisol) has been termed Anxious Temperament (AT); subjects that display BI are higher in AT than their less inhibited counterparts.

Decades of work has shown that the HIP can be used to model BI in young macaques, with extreme responders displaying freezing behaviors and a decrease in vocalizations, similar to childhood BI. Additionally, the context-specific behaviors during NEC begin to emerge around 3 to 4 months of age in rhesus monkeys (Kalin et al., 1991, 1998), a timeframe aligning with the onset of stranger anxiety in children (Brooker et al., 2013). The emergence of context-specific responses during infancy complicates the interpretation of behavior of infants exposed to the NEC context, making this paradigm ideally suited for childhood or adolescent assessment. To this end, Kalin and colleagues have assessed AT in large groups of ~1-3 year old animals (n>500), which is consistent with the peri-adolescent period (Fox et al., 2015, 2018; Shackman et al., 2017). Together these findings highlight the translational relevance of using NEC to assess behavioral inhibition and suggest that the human intruder paradigm is an appropriate model of early life behavioral inhibition in NHPs.

Infant Inhibited temperament (IT)

A complementary measure for reliably assessing BI in infants was developed at the California National Primate Research Center (CNPRC) as a part of the Biobehavioral Assessment (BBA). The BBA is a research program was created in order to characterize an animals’ psychological and physiological

systems, referred to as their “biobehavioral organization” (Capitanio, 2021). This includes characteristics like an animal’s coping abilities, stress reactivity, personality, and temperament. Briefly, the BBA consists of a standardized series of assessments conducted over a 25-hour period in 3-4 month old rhesus monkeys (Capitanio, 2021). The animals' young age enables researchers to evaluate subjects with minimal social exposure, facilitating the assessment of innate, temperamental characteristics unaffected by experiences (Capitanio, 2021). To date, more than 4000 animals have undergone BBA testing.

For most of the animals, enrollment in the BBA involves experiencing changes to their daily routines for the first time. This offers a unique opportunity to identify an inhibited phenotype in the BBA program by seeing how each animal responds and adapts to this change across the 25-hour period (Capitanio, 2021). To this end, the BBA conducts behavioral observations on animals at the beginning (Day 1) and at the end (Day 2) of the 25-hr testing period (Capitanio, 2018). Exploratory and confirmation analyses of behavior during the testing period revealed two latent traits, *Activity* and *Emotionality*. *Activity* is measured as time the animal spent locomoting, time eating or drinking water, and time not hanging from the top or side of their cage, or was observed crouching in the cage. *Emotionality* includes the animal’s rates of vocalizations (cooing; barking), as well as any instances of lip-smacking, displaying threats, or scratching. The BBA’s measure of *inhibited temperament* (IT) is the inverse-average of these two latent variables (i.e. *Activity* and *Emotionality*) across both days. Subjects exhibiting below-average scores on these dimensions on both Day 1 and Day 2 are categorized as behaviorally inhibited, indicating elevated levels of IT. Specifically, animals characterized as having an *inhibited temperament* show lower levels of *Activity* and *Emotionality* on both days of BBA testing. Presumably, this is driven by an inability to adapt to their new surroundings (Capitanio, 2021). Importantly, the BBA finds that ~18% of the total number of animals that have undergone BBA testing fit this measure of infant *inhibited temperament*, a number similar to the 15% of children reported by Kagan (1989) to be BI (Capitanio, 2018). This developmentally-appropriate measure of infant IT provides an opportunity to understand the genesis of BI that can complement peri-adolescent IT measurements and strengthen our inferences about the underlying BI phenotype.

NHP BI is stable and heritable

Though peri-adolescent AT and infant IT are assessed using different behaviors and methodologies, both consistently serve as reliable experimental measures of behavioral inhibition in NHPs. As outlined below, both research programs have found that high-BI animals tend to display inhibition and increased physiological reactivity across potentially-threatening or novel contexts across development, as compared to their low-BI conspecifics. Moreover, similar to human studies (Clauss et al., 2015), individual differences in BI have also been shown to be heritable, suggesting a biological basis for BI.

Like human BI, both peri-adolescent AT and infant IT are stable over time and context. Repeated assessment of peri-adolescent AT has revealed that it is stable over time (Fox et al., 2008, 2019; Shackman et al., 2017). Since BI can often manifest as a latency for rewards, or withdrawals, we hypothesized that high-BI animals would forgo a reward in the presence of a potentially threatening stimulus. Indeed, we found that years after initial assessment, adult animals with high levels of IT during infancy were more likely to refuse treats from a potentially-threatening human intruder (Fox et al., 2021). This data suggests that infant IT is a trait-like measure that is stable across development. Similarly, peri-adolescent high-BI animals were less likely to reach for a treat in the presence of a live snake compared to their less inhibited counterparts (Shackman et al., 2017). These data demonstrate that peri-adolescent AT and infant IT are stable over time and context, indicative of trait-like inhibition and can be used to understand the early phenotypic risk to develop anxiety.

Both peri-adolescent AT and infant IT are significantly heritable, suggesting a genetic component for the development of BI. The first hint of the heritability of BI comes from studies in Wisconsin where

infants and their mothers were shown to have similar behavioral and hormonal responses in response to exposure to the NEC context (Kalin et al., 1998). Both BI measures were collected in large groups of animals at National Primate Research Centers (NPRCs), which are breeding colonies for rhesus monkeys. As such, assessed animals are part of large multi-generational family pedigrees, which provide information about the estimated percentage of genetic similarity between animals. This enables well-powered heritability analyses of phenotypic measures, by determining how phenotypic measures, such as BI, fall through the family tree. Results have demonstrated that both peri-adolescent AT and infant IT to be significantly heritable (Fox et al., 2015, 2021). Importantly, the findings from these studies in primates reveal heritability estimates for AT and IT that are similar to heritability estimates for anxiety disorders in humans (Clauss et al., 2015; Kagan et al., 1988). Overall, this data suggests that both peri-adolescent AT and infant IT are heritable endophenotypes that remain stable across development. As potentially continuous measures, this suggests that elucidating the biological underpinnings of BI is beneficial in identifying early-life interventions and better pharmacological interventions.

Finally, the consistent prediction of peri-adolescent AT and infant IT across various settings underscores their reliability as robust indicators of BI and BI-related traits. This consistent association between AT, IT, and other anxiety-related behaviors suggests that behaviors observed in tasks beyond the laboratory setting can effectively approximate composite BI measures.

The Neurobiology of BI

Early theories about the biological processes of BI proposed that the amygdala would be important in the emergence of BI because of the extensive body of work implicating it in the expression of fear (Blackford & Zald, 2018; Kagan et al., 1988; Kagan & Snidman, 1991). In support of this, the first neuroimaging study of BI found heightened amygdala reactivity in adults humans previously categorized as inhibited (Schwartz et al., 2003). In the decades since, converging evidence has not only supported the importance of the amygdala in BI, but has also revealed evidence of a distributed neural network involved in the expression of BI, trait-like anxiety, and anxiety disorders (Avery et al., 2016; Birn et al., 2014; Etkin et al., 2011; Kenwood et al., 2021; Oler et al., 2012).

A central element within the distributed BI-circuit is the central extended amygdala (EAc), strategically positioned to integrate environmental and interoceptive signals, facilitating adaptive behavioral responses to threatening stimuli (Holley & Fox, 2022). In brief, the EAc, is a functional unit composed of subcortical nuclei along the borders of the amygdala and the ventral striatum, including the central nucleus of the amygdala (Ce) and the bed nucleus of the stria terminalis (BST), as well as portions of the posterior shells of the nucleus accumbens and clusters of cells along the amygdalofugal pathway that connects these regions (often termed sublenticular extended amygdala, or SLEA) (Alheid, 2003; Avery et al., 2016; Shackman & Fox, 2016). The EAc receives multisensory information from different brain regions to coordinate behavioral responses via downstream projections to the periaqueductal gray (PAG), which mediate EAc-induced freezing behavior in mice (Dong & Swanson, 2004; Tovote et al., 2016). In this section, we review the neural substrates identified by IT and AT in the generation and maintenance of BI.

Neural correlates of BI in NHPs

To understand the neural correlates of BI in peri-adolescent monkeys, we have used positron emission tomography (PET) and magnetic resonance imaging (MRI). When used in combination with the NEC paradigm described above, [18F]flourodeoxyglucose PET (FDG-PET) can be used to quantify individual differences in brain metabolism during exposure to the NEC context (i.e. a potential predator).

Subjects are injected with FDG prior to being exposed to the NEC context where they display a range of defensive behaviors. Metabolically active cells take up FDG, which can be quantified in the post-NEC FDG-PET scan to assess circuits involved in BI-relevant behaviors.

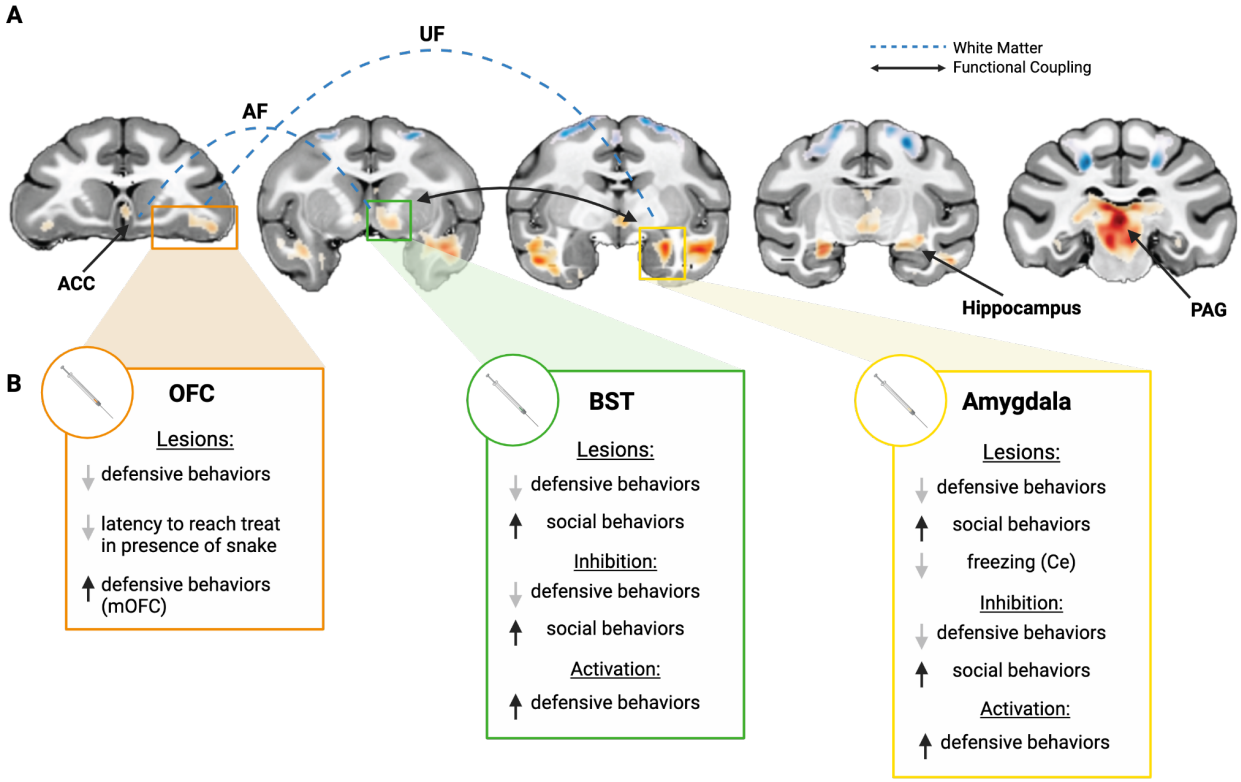

Figure 2: Regions implicated to be associated with NHP BI through neuroimaging (A) and mechanistic studies including lesion and transient approaches (B). The direction of the effect is communicated through arrows. Created with BioRender.com

NEC FDG-PET studies with peri-adolescent animals have rigorously identified AT-related metabolism during exposure to the NEC context across a network of regions including Ce, BST, anterior hippocampus, brainstem, and various PFC regions (Fox et al., 2015; Oler et al., 2010) (**Fig 2**). These findings have been replicated across multiple cohorts and studies, ultimately encompassing a sample of over 600 animals (A. S. Fox et al., 2008; A. S. Fox, Oler, Shackman, et al., 2015; Holley et al., 2024; Oler et al., 2010). These findings underscore a key insight from rhesus monkey FDG-PET studies - there is no singular brain region responsible for the inhibited phenotype; rather, a network of brain regions act together to induce behavioral inhibition (A. S. Fox, Oler, Shackman, et al., 2015). In fact, metabolism in individual brain regions account for a small proportion of the variance in AT on their own (A. S. Fox, Oler, Shackman, et al., 2015). In addition, FDG-PET studies have been used to assess brain metabolism in the absence of direct threats, enabling exploration of trait-like aspects of BI-related metabolism to determine whether BI is due to alterations in the brain's reactivity to the potential threat or baseline differences in brain metabolism. Several studies have shown that when animals are tested during the alone phase of the HIP (i.e. without the direct threat of the human intruder present), increased freezing and decreased coo vocalizations are associated with increased amygdala metabolism (A. S. Fox et al., 2005, 2008; A. S. Fox, Oler, Shackman, et al., 2015; Kalin et al., 2005; Oler et al., 2010). Moreover, peri-adolescent AT assessed during exposure to the NEC context is correlated with increased amygdala metabolism in non-stressful environments, as animals were allowed to freely go about life in their home-cage (Fox et al.,

2008). Additionally, rhesus monkeys with high BI show sustained and increased BST metabolism and freezing during a 30 minute recovery period following exposure to the human intruder in the NEC paradigm, when the threat is no longer present (Shackman et al., 2017). Together, these data suggest that BI is due to both baseline differences in brain regional metabolism and increased glucose metabolism in BI-relevant regions across a variety of threatening contexts.

Heritability analyses have revealed brain metabolism in the majority of AT-relevant brain regions to be significantly heritable, including the BST, Ce, brainstem, and PFC subregions (Fox et al., 2015; Oler et al., 2010). The degree of heritability can vary significantly across regions, with metabolism in the hippocampus and BST showing heritability rates of 53% and 57% respectively, while heritability in Ce metabolism is less than 26% (Fox et al., 2015; Oler et al., 2010). In addition to highlighting the contribution of inherited factors, these results emphasize a key role for environmental factors in BI-related brain metabolism. Critically, heritability of metabolism in AT-related regions does not necessarily implicate these regions in the intergenerational transfer of AT; different genes can both be heritable without sharing an overlapping genetic basis. For example, traits like height and eye color are both heritable, though they are not dependent on the same genes. To identify brain regions where metabolism mediates the intergenerational transfer of AT, we assessed the extent to which brain metabolism and AT fall together through the family tree using genetic correlation analyses. These analyses revealed significant genetic correlations with AT in a network of brain regions that include the OFC, BST, and PAG, suggesting that the genes that alter metabolism in these regions mediate the heritable components of AT (Fox et al., 2015). It is worth noting that the identification of specific regions where metabolism is co-inherited with AT does not necessarily imply that these genes are influencing the function of those cells specifically, rather, it could be that genes influencing activity in one region are responsible for metabolic changes in another.

Functional MRI (fMRI) work can help shed light on the role that interactions between BI-relevant brain regions may play in AT. fMRI BOLD signals measured by "resting state" fMRI under light anesthesia can be used to identify significant correlations in fluctuations in activity across multiple brain regions, allowing researchers to infer connectivity between regions. Within the EAc, resting state connectivity analyses have revealed that the Ce and BST form a functional network with each other; BOLD signal in the Ce is significantly correlated with BOLD signal in the BST (Oler et al., 2012). This strong temporal coupling seems to influence the BI phenotype, as functional connectivity between the Ce and BST is associated with higher AT (A. S. Fox et al., 2018). Importantly, the Ce-BST functional connectivity is heritable and genetically correlated with AT, suggesting a shared genetic substrate (A. S. Fox et al., 2018). These data reinforce the importance of understanding the contribution of the EAc to BI.

Expanding on the network of BI-relevant brain regions identified using FDG-PET methods, fMRI research has also corroborated the role of multiple brain regions in influencing BI, primarily through their connectivity with the EAc. One region of interest is the expanded NHP cortex, where direct and indirect projections with the limbic system play important roles in complex aspects of threat processing. For example, the Ce shows intrinsic functional connectivity with AT-related subregions in the PFC, and increased functional connectivity between the dorsal lateral PFC (dlPFC) and Ce predicts lower levels of AT in young rhesus monkeys (Birn et al., 2014). The mechanism for this connectivity remains unknown and likely involve indirect projections, as there are no robust direct projections between Ce and dlPFC. These data further support a distributed circuit engaging direct and indirect projections to influence AT and highlight a role for corticolimbic coupling in BI.

It is also possible that individual differences in the physical connections between BI-related regions can influence the BI-phenotype. Recent studies in humans have suggested that alterations in the microstructure of the Uncinate Fasciculus (UF), a white matter tract that connects areas of the PFC to the amygdala, may be important in the development of anxiety disorders (Adluru et al., 2017; Baur et al., 2013; Phan et al., 2009; Tromp et al., 2012; Tromp, Williams, et al., 2019). To further understand the contribution of fronto-limbic connections on BI, we used diffusion tensor imaging (DTI) to quantify changes in UF microstructure. In a study of over 500 young rhesus macaques, we found that changes in

the microstructure of the UF was correlated with AT in male, but not female monkeys (Tromp, Fox, et al., 2019). This result complements human studies that found altered UF microstructure in boys with anxiety disorders (Aggarwal et al., 2022;Tromp et al., 2019). Given the well documented sex difference in the prevalence of anxiety disorders, identifying sex-specific mechanisms that underlie BI-related fronto-limbic circuitry is important. Interestingly, in a small preliminary sample, we found that changes in the microstructure of the Accumbofrontal Fasciculus (AF), a white matter tract that connects the PFC to areas of the limbic system, was significantly correlated with IT in a cohort of all female rhesus monkeys (unpublished data). Together, these data suggest the possibility of sexual dimorphism in the network dynamics of the BI phenotype.

Finally, results from the FDG-PET, fMRI, and DTI studies discussed above can complement each other, providing additional insights into how functional connectivity between two brain regions influences downstream brain metabolism. When leveraging these methods together, results show that Ce–BST functional connectivity, while not associated with Ce or BST metabolism, is associated with increased metabolism in shared downstream targets. Specifically, Ce-BST functional connectivity is associated with increased FDG metabolism in AT-related regions including the hypothalamus, SLEA, and PAG (A. S. Fox et al., 2018). These data further highlight the EAc's role in driving BI through a broader network of AT-relevant regions. Additionally, increased dlPFC-CeA connectivity, which is associated with AT, also predicts Ce metabolism (Birn et al., 2014). Together, these data underscore the importance of multimodal imaging techniques to further parse the influences of BI-relevant brain regions on the broader distributed network of BI-related regions.

In conclusion, collective evidence from a diverse array of imaging modalities underscores the intricate network dynamics underlying the expression of BI in peri-adolescent animals. Importantly, these studies consistently reveal that BI is not mediated by any singular brain region but rather by a network of interconnected regions, highlighting the distributed nature of the neural correlates of BI. Similarly, results from heritability analyses demonstrate the complex interplay between genetic predisposition and the environment, further highlighting the complexity of BI. Thus, integrating findings from multimodal imaging techniques will be necessary to provide a comprehensive understanding of how BI-relevant brain regions interact to drive behaviors.

Identification of BI-related molecules in NHPs

Having identified specific brain-regions, researchers have begun to investigate the molecules that might give rise to BI using DNA and RNA sequencing. DNA sequencing is a powerful tool in NHPs, because it can be combined with the large pedigrees at National Primate Research Centers (NPRCs) to perform well-powered genome-wide association studies (GWAS) that leverage pedigree information, much like a large-scale family study (Glahn et al., 2019). Using this approach, we identified a single nucleotide polymorphism (SNP) in a primate-specific exon of the CRHR1 gene to be associated with individual differences in peri-adolescent AT (Rogers et al., 2013). Moreover, by sequencing the whole-genome, we further identified a SNP near the CTNNA2 gene which was associated with infant IT (A. S. Fox et al., 2021b). These studies each identify a specific molecular process that is sufficient to give rise to BI, and are consistent with human research focusing on BI-related phenotypes and psychopathology (Gonda et al., 2021; Lesch et al., 2008; Smith et al., 2016). For example, genetic association studies of neuroticism and anxiety-related disorders in humans support our data implicating CRHR1 (Levey et al., 2020; Nagel et al., 2018; Stein et al., 2021). These data further support the NHP as a model organism for discovery, which can be further leveraged by the robust genetic variation in NHPs. Nevertheless, it is important to note that these studies are merely scratching the surface of the specific molecules that might give rise to the genesis of BI and more work needs to be done.

In addition to understanding the structural genetic changes that give rise to BI, another line of work has focused on RNA expression, searching for transcripts associated with individual differences in BI. This work has identified numerous molecules which are associated with variation in BI, including NTRK3, which was replicated in two independent samples of >20 animals (A. S. Fox et al., 2012, 2019). NTRK3 is a growth factor receptor, similar to the NTRK2 receptor, which binds BDNF, a gene that has been repeatedly implicated by rodent models of anxiety and depression (Amador-Arjona et al., 2010; Duman, 2017; Farhang et al., 2014). In another gene expression study focusing on CeL neurons, we identified AT-related expression of additional genes including PRKCD, which was positively correlated with AT (Kovner et al., 2020). PRKCD-expressing neurons are particularly interesting because of the role that they play in rodent freezing behavior (Beyeler & Dabrowska, 2020; Fadok et al., 2018; Haubensak et al., 2010). Interestingly, we found that PRKCD expressing neurons represented the majority of BST-projecting CeL cells, highlighting a potential role in mediating Ce-BST functional connectivity outlined above. Similar studies have been undertaken in posterior OFC (consistent with the location of the AT-related metabolism in OFC), where we identified layer-specific AT-related expression in genes that included the gene that encodes the glucocorticoid receptor NR3C1. Each of these discovery-based studies, begin to elucidate the molecular basis of BI, and the stage for mechanistic interventions that specifically target particular molecules to assess their causal role in altering BI.

Targeted manipulations of BI-related regions

NHP lesion studies have begun to uncover the causal contributions of BI-related regions to threat processing and social behavior. For instance, lesions encompassing the whole amygdala are sufficient to decrease an animal's latency to retrieve a food-reward in the presence of a snake -- the opposite of what we see in animals with high-levels of peri-adolescent AT (Chudasama et al., 2009; Izquierdo et al., 2005; Kalin et al., 2001, 2004). Amygdala lesions also decrease freezing and other defensive responses, and are sufficient to decreased social reticence (Amaral, 2002, though, c.f. Kalin et al., 2001). Moreover, Ce-specific lesions, which spare the majority of the amygdala including fibers of passage, are sufficient to decrease freezing and increase the frequency of coo vocalizations (Kalin et al., 2004). Again, these results are consistent with the neuroimaging data, and result in changes that are the opposite of what has been observed in high-BI animals. Critically, although lesions of the amygdala reduce BI, they do not abolish it. This is consistent with multiple brain circuits each contributing to an underlying inhibited temperament.

Another region that has been shown to causally contribute to BI is the OFC. Targeted lesions to the OFC have been shown to mediate aspects of threat processing. Importantly, this large and heterogeneous area of the cortex is structurally connected to the amygdala via the UF (Folloni et al., 2019; Price, 2007; Wallis, 2012).Similar to lesions of the amygdala, lesioning the OFC decreases freezing behaviors (A. Fox et al., 2010; Izquierdo et al., 2005; Kalin et al., 2007; Machado & Bachevalier, 2008; Murray & Izquierdo, 2007). Additionally, rhesus monkeys with OFC lesions or unilateral OFC and amygdala lesions show shorter latencies to reach for a preferred treat in the presence of a snake-like object (Kalin et al., 2007; Murray & Izquierdo, 2007). However, monkeys with medial OFC lesions were also shown to have increased defensive behaviors to both threatening and non-threatening stimuli (Pujara et al., 2019). More recently, Kenwood et al. (2019) found that so-called "strip" lesions to the OFC, which spare most OFC gray-matter but damage fibers in the UF and AF, are sufficient to decrease freezing and BST metabolism during “no eye contact” (NEC). Overall, these studies demonstrate a causal role for OFC in threat-processing, and suggest that the OFC could contribute to the maladaptive regulation of anxiety and BI via projections to the BST and amygdala.

While lesion studies have supported the role of the amygdala and OFC in threat processing, results have been mixed due to inadvertent damage to nearby neuronal projection fibers or fibers passing

through these regions, rather than the deficits being dependent on amygdala and OFC (Rudebeck et al., 2013). In support of this idea, excitotoxic lesions of the OFC, which spare fibers of passage, did not show changes in behavior during snake fear (Rudebeck et al., 2013). However, a follow-up study using excitotoxic lesions to the medial OFC (Walker's areas 14) saw increases in defensive behaviors (Pujara et al., 2019). Notably, this is in direct contrast to previous lesions to the OFC that found decreases in defensive responses (A. Fox et al., 2010; Izquierdo et al., 2005; Kalin et al., 2007; Machado & Bachevalier, 2008).These findings underscore the importance of understanding precisely what regions and fibers are damaged when interpreting the results of lesion studies. Since both the OFC and amygdala are heterogeneous, the exact location of lesions may perturb different circuit-dynamics within these regions, as projections to and from these regions tend to be topographically organized (Price, 2007; Saleem et al., 2008; Swanson & Petrovich, 1998). Further research using more precise targeting strategies will provide additional insight into the role of these regions in anxiety like behaviors and BI.

Another limitation of lesion experiments is their irreversibility, which allows for compensatory and indirect mechanisms to influence the observed effects. Efforts are ongoing to understand the specific circuit dynamics of these BI-related regions using more targeted and transient techniques. Consistent with the findings of lesion studies, transient activation and inactivation studies have demonstrated a causal role for amygdala and frontal regions in BI-related behaviors. For example, inhibition of the rhesus monkey amygdala using chemogenetics was also sufficient to decrease anxiety-like freezing during exposure to the NEC context (Roseboom et al., 2021). More recently, chemogenetic activation of the amygdala increased freezing, even in inappropriate contexts such as when a human intruder is staring at the animal (Mueller et al., 2023). Relatedly, transient inactivation of the BST (Jacobs et al., 2023) and Ce (Wellman et al., 2016) was sufficient to increase affiliative behaviors (e.g. grooming) towards a familiar conspecific in macaque monkeys. In marmosets, pharmacological overactivation of the ACC (Area 25)

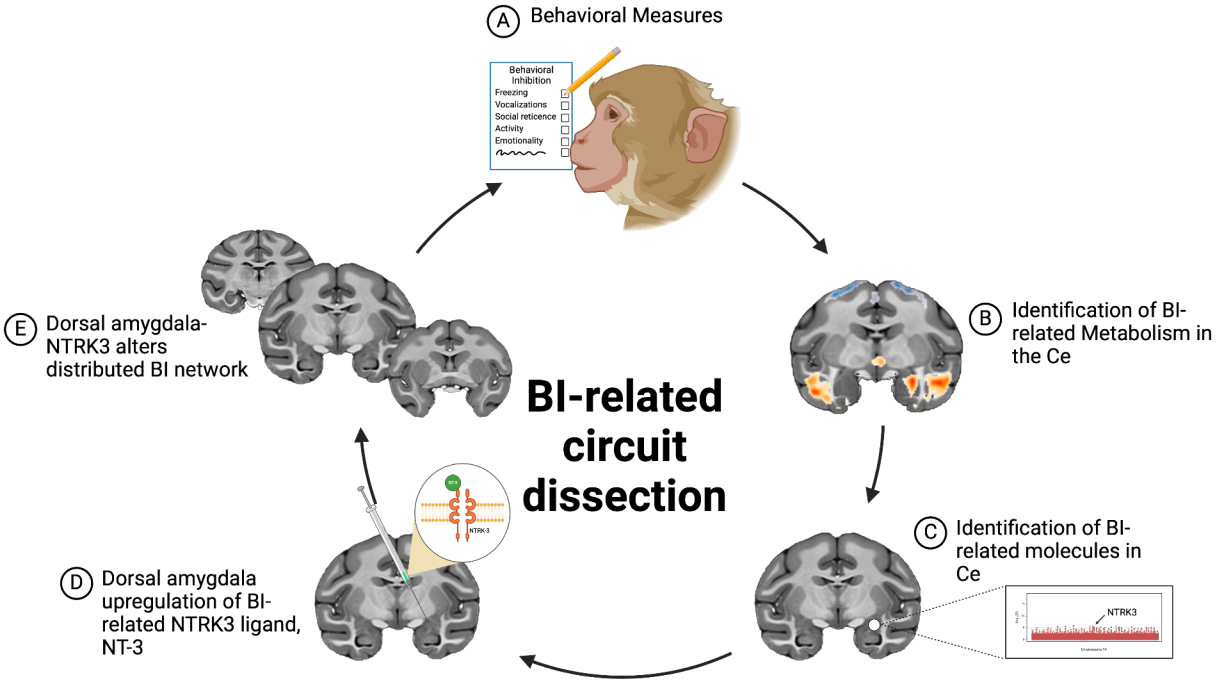

Figure 3: Overview of Fox et al., 2019 highlights how NHP models can be used to dissect BI-related circuits using different levels of analysis. Specifically, NHPs can help (A) identify BI-related behaviors that are (B) correlated with specific circuits, (C) identify BI-related molecules in those circuits (D), which can then be targeted for viral-mediated manipulations (E), to examining their impact on distributed BI-related circuits (E), and behavior (A). Created with BioRender.com.

heightened anxiety-like behaviors in response to an unknown human (Roberts & Clarke, 2019). In marmoset OFC, pharmacological inactivation increased anxiety-like behavior during a human intruder test (Alexander et al., 2020; Stawicka et al., 2022), mirroring the results of lesion studies in rhesus monkeys (Pujara et al., 2019). Together, these studies show that temporally-specific manipulations of amygdala and OFC are sufficient to alter BI, opening the door to a more precise understanding of how these regions contribute to BI **(Fig 3)**.

Using similar techniques, researchers have begun to test the causality of some of the specific molecules (identified above) within these regions to alter BI. For example, using a viral vector to overexpress CRH, the ligand that binds to CRHR1, we were able to show that increasing CRH in the Ce-containing dorsal amygdala region is sufficient to increase peri-adolescent AT (Kalin et al., 2016). In contrast, overexpressing NTF3, the ligand that binds to NTRK3, was sufficient to decrease peri-adolescent AT (Fox et al., 2019). These studies demonstrate that altering a specific molecule in the dorsal amygdala is sufficient to produce large and meaningful alterations in BI. Critically, each of these studies leveraged neuroimaging to understand the impact of specific molecular interventions on the distributed AT circuit. Specifically, molecular manipulations in dorsal amygdala were sufficient to impact metabolism across the AT-network in regions including hippocampus, BST, and OFC (Fox et al., 2019). These results demonstrate that localized manipulation of a single molecule within a single region is sufficient to alter the function of a distributed neural circuit, underscoring the interconnections across the distributed network of brain regions that contribute to BI. Although this work is in its infancy, and will require substantive development of molecular tools for optimal manipulations of NHP brain circuits (e.g. Campos et al., 2023) these results highlight the promise of defining, monitoring, and manipulating specific regional molecular systems in NHPs.

Conclusion

Here we have introduced two rhesus monkey models of BI focused on distinct developmental periods, and outlined how these models can be leveraged to better understand the neurobiology of BI. As in humans, studies in rhesus monkeys have implicated a distributed brain network that includes the EAc in early-life BI. In addition to solidifying NHP models of BI, these studies complement human studies by implicating specific neural substrates in the intergenerational transmission of BI, suggesting that genes that affect metabolism in the BST and Ce-BST connectivity contribute to BI. Identifying the regions that contribute to BI has enabled researchers to begin to uncover the molecular underpinnings of BI using DNA and RNA-sequencing, such as *CRHR1* and *NTRK3*. Furthermore, the NHP model enables manipulation of brain regions and specific molecules to elucidate their causal contribution to BI. This work suggests that OFC may influence BI via projections to the BST, that the Ce causally contributes to BI, and demonstrated that manipulations of specific molecular pathways in the Ce, including CRH and NTF-3, are sufficient to alter BI. Importantly, this work shows that manipulating a single molecule in a specific brain region is sufficient to alter function throughout the distributed network of BI-related brain regions. Despite this progress, there remain several challenges for NHP research to bridge the gap between rodent and human studies, including tools for viral-mediated transfection and cell-type-specific manipulations. This work is ongoing, and promises to aid in the dissection of these BI-related circuits (Chen et al., 2022, 2023; Chuapoco et al., 2023). Together, these data shed light on the neurobiology that underlies the risk to develop anxiety and depressive disorders, and underscore the utility of NHP models for understanding BI. Ultimately, a refined understanding of BI's neurobiology promises to guide the development of new strategies for treating and preventing debilitating psychopathology.

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References

Aggarwal, N., Williams, L. E., Tromp, D. P. M., Pine, D. S., & Kalin, N. H. (2022). A dynamic relation between whole-brain white matter microstructural integrity and anxiety symptoms in preadolescent females with pathological anxiety. *Translational Psychiatry*, *12*(1), 57. https://doi.org/10.1038/s41398-022-01827-y

Adluru, N., Luo, Z., Van Hulle, C. A., Schoen, A. J., Davidson, R. J., Alexander, A. L., & Goldsmith, H. H. (2017). Anxiety-related experience-dependent white matter structural differences in adolescence: A monozygotic twin difference approach. *Scientific Reports*, *7*. https://doi.org/10.1038/s41598-017-08107-6

Alexander, L., Wood, C. M., Gaskin, P. L. R., Sawiak, S. J., Fryer, T. D., Hong, Y. T., McIver, L., Clarke, H. F., & Roberts, A. C. (2020). Over-activation of primate subgenual cingulate cortex enhances the cardiovascular, behavioral and neural responses to threat. *Nature Communications*, *11*(1), 5386. https://doi.org/10.1038/s41467-020-19167-0

Alheid, G. F. (2003). Extended amygdala and basal forebrain. *Annals of the New York Academy of Sciences*, *985*, 185–205.

Amador-Arjona, A., Delgado-Morales, R., Belda, X., Gagliano, H., Gallego, X., Keck, M. E., Armario, A., & Dierssen, M. (2010). Susceptibility to stress in transgenic mice overexpressing TrkC, a model of panic disorder. *Journal of Psychiatric Research*, *44*(3), 157–167. https://doi.org/10.1016/j.jpsychires.2009.07.008

Amaral, D. G. (2002). The Primate Amygdala and the Neurobiology of Social Behavior: Implications for Understanding Social Anxiety. *Biological Psychiatry*, *51*, 11–17.

Avery, S. N., Clauss, J. A., & Blackford, J. U. (2016). The Human BNST: Functional Role in Anxiety and Addiction. *Neuropsychopharmacology*, *41*(1), 126–141. https://doi.org/10.1038/npp.2015.185

Azkona, G., & Sanchez-Pernaute, R. (2022). Mice in translational neuroscience: What R we

doing? *Progress in Neurobiology*, *217*, 102330. https://doi.org/10.1016/j.pneurobio.2022.102330

Baur, V., Brühl, A. B., Herwig, U., Eberle, T., Rufer, M., Delsignore, A., Jäncke, L., & Hänggi, J. (2013). Evidence of frontotemporal structural hypoconnectivity in social anxiety disorder: A quantitative fiber tractography study. *Human Brain Mapping*, *34*(2), 437–446. https://doi.org/10.1002/hbm.21447

Belmonte, J. C. I., Callaway, E. M., Caddick, S. J., Churchland, P., Feng, G., Homanics, G. E., Lee, K.-F., Leopold, D. A., Miller, C. T., Mitchell, J. F., Mitalipov, S., Moutri, A. R., Movshon, J. A., Okano, H., Reynolds, J. H., Ringach, D. L., Sejnowski, T. J., Silva, A. C., Strick, P. L., … Zhang, F. (2015). Brains, Genes, and Primates. *Neuron*, *86*(3), 617–631. https://doi.org/10.1016/j.neuron.2015.03.021

Beyeler, A., & Dabrowska, J. (2020). Neuronal diversity of the amygdala and the bed nucleus of the stria terminalis. In *Handbook of Behavioral Neuroscience* (Vol. 26, pp. 63–100). Elsevier. https://doi.org/10.1016/B978-0-12-815134-1.00003-9

Biederman, J., Hirshfeld-Becker, D. R., Rosenbaum, J. F., Hérot, C., Friedman, D., Snidman, N., Kagan, J., & Faraone, S. V. (2001). Further evidence of association between behavioral inhibition and social anxiety in children. *The American Journal of Psychiatry*, *158*(10), 1673–1679. https://doi.org/10.1176/appi.ajp.158.10.1673

Birn, R. M., Shackman, A. J., Oler, J. A., Williams, L. E., McFarlin, D. R., Rogers, G. M., Shelton, S. E., Alexander, A. L., Pine, D. S., Slattery, M. J., Davidson, R. J., Fox, A. S., & Kalin, N. H. (2014). Evolutionarily conserved prefrontal-amygdalar dysfunction in early-life anxiety. *Molecular Psychiatry*, *19*(8), 915–922. https://doi.org/10.1038/mp.2014.46

Blackford, J. U., & Zald, D. H. (2018). Inhibited temperament and intrinsic versus extrinsic influences on fear circuits. In *The Nature of Emotion: Fundamental Questions* (Second Edition, pp. 49–50). Oxford University Press.

Brooker, R. J., Buss, K. A., Lemery-Chalfant, K., Aksan, N., Davidson, R. J., & Goldsmith, H. H.

(2013). The development of stranger fear in infancy and toddlerhood: Normative development, individual differences, antecedents, and outcomes. *Developmental Science*, *16*(6), 864–878. https://doi.org/10.1111/desc.12058

Campos, L. J., Arokiaraj, C. M., Chuapoco, M. R., Chen, X., Goeden, N., Gradinaru, V., & Fox, A. S. (2023). Advances in AAV technology for delivering genetically encoded cargo to the nonhuman primate nervous system. *Current Research in Neurobiology*, *4*, 100086. https://doi.org/10.1016/j.crneur.2023.100086

Capitanio, J. P. (2018). Behavioral Inhibition in Nonhuman Primates: The Elephant in the Room. In K. Perez-Edgar & N. A. Fox (Eds.), *Behavioral Inhibition: Integrating Theory, Research, and Clinical Perspectives* (pp. 17–33). Springer Nature Switzerland AG.

Capitanio, J. P. (2021). Knowledge of Biobehavioral Organization Can Facilitate Better Science: A Review of the BioBehavioral Assessment Program at the California National Primate Research Center. *Animals*, *11*(8), 2445. https://doi.org/10.3390/ani11082445

Cavigelli, S. A., & McClintock, M. K. (2003). Fear of novelty in infant rats predicts adult corticosterone dynamics and an early death. *Proceedings of the National Academy of Sciences*, *100*(26), 16131–16136. https://doi.org/10.1073/pnas.2535721100

Chang, S. W. C., & Platt, M. L. (2014). Oxytocin and social cognition in rhesus macaques: Implications for understanding and treating human psychopathology. *Brain Research*, *1580*, 57–68. https://doi.org/10.1016/j.brainres.2013.11.006

Chen, X., Ravindra Kumar, S., Adams, C. D., Yang, D., Wang, T., Wolfe, D. A., Arokiaraj, C. M., Ngo, V., Campos, L. J., Griffiths, J. A., Ichiki, T., Mazmanian, S. K., Osborne, P. B., Keast, J. R., Miller, C. T., Fox, A. S., Chiu, I. M., & Gradinaru, V. (2022). Engineered AAVs for non-invasive gene delivery to rodent and non-human primate nervous systems. *Neuron*, *110*(14), 2242-2257.e6. https://doi.org/10.1016/j.neuron.2022.05.003

Chen, X., Wolfe, D. A., Bindu, D. S., Zhang, M., Taskin, N., Goertsen, D., Shay, T. F., Sullivan, E. E., Huang, S.-F., Ravindra Kumar, S., Arokiaraj, C. M., Plattner, V. M., Campos, L. J.,

Mich, J. K., Monet, D., Ngo, V., Ding, X., Omstead, V., Weed, N., … Gradinaru, V. (2023). Functional gene delivery to and across brain vasculature of systemic AAVs with endothelial-specific tropism in rodents and broad tropism in primates. *Nature Communications*, *14*(1), 3345. https://doi.org/10.1038/s41467-023-38582-7

Chuapoco, M. R., Flytzanis, N. C., Goeden, N., Christopher Octeau, J., Roxas, K. M., Chan, K. Y., Scherrer, J., Winchester, J., Blackburn, R. J., Campos, L. J., Man, K. N. M., Sun, J., Chen, X., Lefevre, A., Singh, V. P., Arokiaraj, C. M., Shay, T. F., Vendemiatti, J., Jang, M. J., … Gradinaru, V. (2023). Adeno-associated viral vectors for functional intravenous gene transfer throughout the non-human primate brain. *Nature Nanotechnology*. https://doi.org/10.1038/s41565-023-01419-x

Chudasama, Y., Izquierdo, A., & Murray, E. A. (2009). Distinct contributions of the amygdala and hippocampus to fear expression. *European Journal of Neuroscience*, *30*(12), 2327–2337. https://doi.org/10.1111/j.1460-9568.2009.07012.x

Chun, K., & Capitanio, J. P. (2016). Developmental consequences of behavioral inhibition: A model in rhesus monkeys (Macaca mulatta). *Developmental Science*, *19*(6), 1035–1048. https://doi.org/10.1111/desc.12339

Clauss, J. A., Avery, S. N., & Blackford, J. U. (2015). The nature of individual differences in inhibited temperament and risk for psychiatric disease: A review and meta-analysis. *Progress in Neurobiology*, *127–128*, 23–45. https://doi.org/10.1016/j.pneurobio.2015.03.001

Clauss, J. A., & Blackford, J. U. (2012). Behavioral inhibition and risk for developing social anxiety disorder: A meta-analytic study. *Journal of the American Academy of Child and Adolescent Psychiatry*, *51*(10), 1066-1075.e1. https://doi.org/10.1016/j.jaac.2012.08.002

Clauss, J. A., Seay, A. L., VanDerKlok, R. M., Avery, S. N., Cao, A., Cowan, R. L., Benningfield, M. M., & Blackford, J. U. (2014). Structural and functional bases of inhibited temperament. *Social Cognitive and Affective Neuroscience*, *9*(12), 2049–2058.

https://doi.org/10.1093/scan/nsu019

Dong, H., & Swanson, L. W. (2004). Projections from bed nuclei of the stria terminalis, posterior division: Implications for cerebral hemisphere regulation of defensive and reproductive behaviors. *Journal of Comparative Neurology*, *471*(4), 396–433. https://doi.org/10.1002/cne.20002

Duman, R. S. (2017). BDNF, 5-HT, and Anxiety: Identification of a Critical Periadolescent Developmental Period. *American Journal of Psychiatry*, *174*(12), 1137–1139. https://doi.org/10.1176/appi.ajp.2017.17101084

Dunbar, R. I. M., & Shultz, S. (2007). Evolution in the Social Brain. *Science*, *317*(5843), 1344–1347. https://doi.org/10.1126/science.1145463

Etkin, A., Egner, T., & Kalisch, R. (2011). Emotional processing in anterior cingulate and medial prefrontal cortex. *Trends in Cognitive Sciences*, *15*(2), 85–93. https://doi.org/10.1016/j.tics.2010.11.004

Fadok, J. P., Markovic, M., Tovote, P., & Lüthi, A. (2018). New perspectives on central amygdala function. *Current Opinion in Neurobiology*, *49*, 141–147. https://doi.org/10.1016/j.conb.2018.02.009

Farhang, S., Barar, J., Fakhari, A., Mesgariabbasi, M., Khani, S., Omidi, Y., & Farnam, A. (2014). Asymmetrical expression of BDNF and NTRK3 genes in frontoparietal cortex of stress-resilient rats in an animal model of depression. *Synapse*, *68*(9), 387–393. https://doi.org/10.1002/syn.21746

Folloni, D., Sallet, J., Khrapitchev, A. A., Sibson, N., Verhagen, L., & Mars, R. B. (2019). Dichotomous organization of amygdala/temporal-prefrontal bundles in both humans and monkeys. *eLife*, *8*, e47175. https://doi.org/10.7554/eLife.47175

Fox, A. S. (2018). The brain is organized to emote. In *The Nature of Emotions: Fundamental Questions* (Second Edition, pp. 118–121). Oxford University Press.

Fox, A. S., Harris, R. A., Rosso, L. D., Raveendran, M., Kamboj, S., Kinnally, E. L., Capitanio, J.

P., & Rogers, J. (2021). Infant inhibited temperament in primates predicts adult behavior, is heritable, and is associated with anxiety-relevant genetic variation. *Molecular Psychiatry*, *26*(11), 6609–6618. https://doi.org/10.1038/s41380-021-01156-4

Fox, A. S., & Kalin, N. H. (2014). A translational neuroscience approach to understanding the development of social anxiety disorder and its pathophysiology. *The American Journal of Psychiatry*, *171*(11), 1162–1173. https://doi.org/10.1176/appi.ajp.2014.14040449

Fox, A. S., Oakes, T. R., Shelton, S. E., Converse, A. K., Davidson, R. J., & Kalin, N. H. (2005). Calling for help is independently modulated by brain systems underlying goal-directed behavior and threat perception. *Proceedings of the National Academy of Sciences of the United States of America*, *102*(11), 4176–4179. https://doi.org/10.1073/pnas.0409470102

Fox, A. S., Oler, J. A., Birn, R. M., Shackman, A. J., Alexander, A. L., & Kalin, N. H. (2018). Functional Connectivity within the Primate Extended Amygdala Is Heritable and Associated with Early-Life Anxious Temperament. *The Journal of Neuroscience*, *38*(35), 7611–7621. https://doi.org/10.1523/JNEUROSCI.0102-18.2018

Fox, A. S., Oler, J. A., Shackman, A. J., Shelton, S. E., Raveendran, M., McKay, D. R., Converse, A. K., Alexander, A., Davidson, R. J., Blangero, J., Rogers, J., & Kalin, N. H. (2015). Intergenerational neural mediators of early-life anxious temperament. *Proceedings of the National Academy of Sciences*, *112*(29), 9118–9122. https://doi.org/10.1073/pnas.1508593112

Fox, A. S., Oler, J. A., Shelton, S. E., Nanda, S. A., Davidson, R. J., Roseboom, P. H., & Kalin, N. H. (2012). Central amygdala nucleus (Ce) gene expression linked to increased trait-like Ce metabolism and anxious temperament in young primates. *Proceedings of the National Academy of Sciences of the United States of America*, *109*(44), 18108–18113. https://doi.org/10.1073/pnas.1206723109

Fox, A. S., Oler, J. A., Tromp, D. P. M., Fudge, J. L., & Kalin, N. H. (2015). Extending the

amygdala in theories of threat processing. *Trends in Neurosciences*, *38*(5), 319–329. https://doi.org/10.1016/j.tins.2015.03.002

Fox, A. S., & Shackman, A. J. (2019). The central extended amygdala in fear and anxiety: Closing the gap between mechanistic and neuroimaging research. *Neuroscience Letters*, *693*, 58–67. https://doi.org/10.1016/j.neulet.2017.11.056

Fox, A. S., Shelton, S. E., Oakes, T. R., Davidson, R. J., & Kalin, N. H. (2008). Trait-Like Brain Activity during Adolescence Predicts Anxious Temperament in Primates. *PLoS ONE*, *3*(7), e2570. https://doi.org/10.1371/journal.pone.0002570

Fox, A. S., Souaiaia, T., Oler, J. A., Kovner, R., Kim, J. M. (Hugo), Nguyen, J., French, D. A., Riedel, M. K., Fekete, E. M., Rabska, M. R., Olsen, M. E., Brodsky, E. K., Alexander, A. L., Block, W. F., Roseboom, P. H., Knowles, J. A., & Kalin, N. H. (2019). Dorsal Amygdala Neurotrophin-3 Decreases Anxious Temperament in Primates. *Biological Psychiatry*, *86*(12), 881–889. https://doi.org/10.1016/j.biopsych.2019.06.022

Fox, A., Shelton, S., Oakes, T., Converse, A., Davidson, R., & Kalin, N. (2010). Orbitofrontal Cortex Lesions Alter Anxiety-Related Activity in the Primate Bed Nucleus of Stria Terminalis. *The Journal of Neuroscience : The Official Journal of the Society for Neuroscience*, *30*(20), 7023–7027. https://doi.org/10.1523/JNEUROSCI.5952-09.2010

Fox, N. A., Buzzell, G. A., Morales, S., Valadez, E. A., Wilson, M., & Henderson, H. A. (2021). Understanding the Emergence of Social Anxiety in Children With Behavioral Inhibition. *Biological Psychiatry*, *89*(7), 681–689. https://doi.org/10.1016/j.biopsych.2020.10.004

Fox, N. A., Henderson, H. A., Marshall, P. J., Nichols, K. E., & Ghera, M. M. (2005). Behavioral Inhibition: Linking Biology and Behavior within a Developmental Framework. *Annual Review of Psychology*, *56*(1), 235–262. https://doi.org/10.1146/annurev.psych.55.090902.141532

Glahn, D. C., Nimgaonkar, V. L., Raventós, H., Contreras, J., McIntosh, A. M., Thomson, P. A., Jablensky, A., McCarthy, N. S., Charlesworth, J. C., Blackburn, N. B., Peralta, J. M.,

Knowles, E. E. M., Mathias, S. R., Ament, S. A., McMahon, F. J., Gur, R. C., Bucan, M., Curran, J. E., Almasy, L., … Blangero, J. (2019). Rediscovering the value of families for psychiatric genetics research. *Molecular Psychiatry*, *24*(4), 523–535. https://doi.org/10.1038/s41380-018-0073-x

Gonda, X., Eszlari, N., Torok, D., Gal, Z., Bokor, J., Millinghoffer, A., Baksa, D., Petschner, P., Antal, P., Breen, G., Juhasz, G., & Bagdy, G. (2021). Genetic underpinnings of affective temperaments: A pilot GWAS investigation identifies a new genome-wide significant SNP for anxious temperament in ADGRB3 gene. *Translational Psychiatry*, *11*(1), 337. https://doi.org/10.1038/s41398-021-01436-1

Haubensak, W., Kunwar, P. S., Cai, H., Ciocchi, S., Wall, N. R., Ponnusamy, R., Biag, J., Dong, H.-W., Deisseroth, K., Callaway, E. M., Fanselow, M. S., Lüthi, A., & Anderson, D. J. (2010). Genetic dissection of an amygdala microcircuit that gates conditioned fear. *Nature*, *468*(7321), 270–276. https://doi.org/10.1038/nature09553

Holley, D., Campos, L. J., Drzewiecki, C. M., Zhang, Y., Capitanio, J. P., & Fox, A. S. (2024). Rhesus infant nervous temperament predicts peri-adolescent central amygdala metabolism & behavioral inhibition measured by a machine-learning approach. *Translational Psychiatry*, *14*(1), 1–11. https://doi.org/10.1038/s41398-024-02858-3

Holley, D., & Fox, A. S. (2022). The central extended amygdala guides survival-relevant tradeoffs: Implications for understanding common psychiatric disorders. *Neuroscience & Biobehavioral Reviews*, *142*, 104879. https://doi.org/10.1016/j.neubiorev.2022.104879

Izquierdo, A., Suda, R. K., & Murray, E. A. (2005). Comparison of the Effects of Bilateral Orbital Prefrontal Cortex Lesions and Amygdala Lesions on Emotional Responses in Rhesus Monkeys. *The Journal of Neuroscience*, *25*(37), 8534–8542. https://doi.org/10.1523/JNEUROSCI.1232-05.2005

Jacobs, J. T., Maior, R. S., Waguespack, H. F., Campos-Rodriguez, C., Forcelli, P. A., & Malkova, L. (2023). Pharmacological Inactivation of the Bed Nucleus of the Stria

Terminalis Increases Affiliative Social Behavior in Rhesus Macaques. *The Journal of Neuroscience*, *43*(18), 3331–3338. https://doi.org/10.1523/JNEUROSCI.2090-22.2023

Kagan, J., Reznick, J. S., Clarke, C., & Snidman, N. (1984). Behavioral Inhibition to the Unfamiliar. *Child Development*, *55*(6), 2212–2225.

Kagan, J., Reznick, J. S., & Snidman, N. (1987). The Physiology and Psychology of Behavioral Inhibition in Children. *Child Development*, *58*(6), 1459–1473.

Kagan, J., Reznick, J. S., & Snidman, N. (1988). Biological Bases of Childhood Shyness. *Science*, *240*(4849), 167–171. https://doi.org/10.1126/science.3353713

Kagan, J., & Snidman, N. (1991). Temperamental Factors in Human Development. *American Psychologist*, *46*(856–862).

Kalin, N. H., Fox, A. S., Kovner, R., Riedel, M. K., Fekete, E. M., Roseboom, P. H., Tromp, D. P. M., Grabow, B. P., Olsen, M. E., Brodsky, E. K., McFarlin, D. R., Alexander, A. L., Emborg, M. E., Block, W. F., Fudge, J. L., & Oler, J. A. (2016). Overexpressing corticotropin-releasing hormone (CRH) in the primate amygdala increases anxious temperament and alters its neural circuit. *Biological Psychiatry*, *80*(5), 345–355. https://doi.org/10.1016/j.biopsych.2016.01.010

Kalin, N. H., Larson, C., & Shelton, S. E. (1998). Asymmetric Frontal Brain Activity, Cortisol, and Behavior Associated With Fearful Temperament in Rhesus Monkeys. *Behavioral Neuroscience*, *112*(2), 286–292.

Kalin, N. H., & Shelton, S. E. (1989). Defensive behaviors in infant rhesus monkeys: Environmental cues and neurochemical regulation. *Science (New York, N.Y.)*, *243*(4899), 1718–1721.

Kalin, N. H., & Shelton, S. E. (2003). Nonhuman Primate Models to Study Anxiety, Emotion Regulation, and Psychopathology. *Annals of the New York Academy of Sciences*, *1008*(1), 189–200. https://doi.org/10.1196/annals.1301.021

Kalin, N. H., Shelton, S. E., & Davidson, R. J. (2004). The role of the central nucleus of the

amygdala in mediating fear and anxiety in the primate. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, *24*(24), 5506–5515. https://doi.org/10.1523/JNEUROSCI.0292-04.2004

Kalin, N. H., Shelton, S. E., & Davidson, R. J. (2007). Role of the primate orbitofrontal cortex in mediating anxious temperament. *Biological Psychiatry*, *62*(10), 1134–1139. https://doi.org/10.1016/j.biopsych.2007.04.004

Kalin, N. H., Shelton, S. E., Davidson, R. J., & Kelley, A. E. (2001). The Primate Amygdala Mediates Acute Fear But Not the Behavioral and Physiological Components of Anxious Temperament. *The Journal of Neuroscience*, *21*(6), 2067–2074. https://doi.org/10.1523/JNEUROSCI.21-06-02067.2001

Kalin, N. H., Shelton, S. E., Fox, A. S., Oakes, T. R., & Davidson, R. J. (2005). Brain Regions Associated with the Expression and Contextual Regulation of Anxiety in Primates. *Biological Psychiatry*, *58*(10), 796–804. https://doi.org/10.1016/j.biopsych.2005.05.021

Kalin, N. H., Shelton, S. E., & Takahashi, L. K. (1991). Defensive Behaviors in Infant Rhesus Monkeys: Ontogeny and Context-Dependent Selective Expression. *Child Development*, *62*(5), 1175–1183.

Kamath, T., Abdulraouf, A., Burris, S. J., Langlieb, J., Gazestani, V., Nadaf, N. M., Balderrama, K., Vanderburg, C., & Macosko, E. Z. (2022). Single-cell genomic profiling of human dopamine neurons identifies a population that selectively degenerates in Parkinson's disease. *Nature Neuroscience*, *25*(5), 588–595. https://doi.org/10.1038/s41593-022-01061-1

Kenwood, M. M., Kalin, N. H., & Barbas, H. (2021). The prefrontal cortex, pathological anxiety, and anxiety disorders. *Neuropsychopharmacology*. https://doi.org/10.1038/s41386-021-01109-z

Kenwood, M. M., Oler, J. A., Tromp, D. P. M., Fox, A. S., Riedel, M. K., Roseboom, P. H., Brunner, K. G., Aggarwal, N., Murray, E. A., & Kalin, N. H. (2023). Prefrontal influences

on the function of the neural circuitry underlying anxious temperament in primates. *Oxford Open Neuroscience*, *2*, kvac016. https://doi.org/10.1093/oons/kvac016

Kovner, R., Souaiaia, T., Fox, A. S., French, D. A., Goss, C. E., Roseboom, P. H., Oler, J. A., Riedel, M. K., Fekete, E. M., Fudge, J. L., Knowles, J. A., & Kalin, N. H. (2020). Transcriptional Profiling of Primate Central Nucleus of the Amygdala Neurons to Understand the Molecular Underpinnings of Early-Life Anxious Temperament. *Biological Psychiatry*, *88*(8), 638–648. https://doi.org/10.1016/j.biopsych.2020.05.009

Krienen, F. M., Goldman, M., Zhang, Q., C. H. del Rosario, R., Florio, M., Machold, R., Saunders, A., Levandowski, K., Zaniewski, H., Schuman, B., Wu, C., Lutservitz, A., Mullally, C. D., Reed, N., Bien, E., Bortolin, L., Fernandez-Otero, M., Lin, J. D., Wysoker, A., … McCarroll, S. A. (2020). Innovations present in the primate interneuron repertoire. *Nature*, *586*(7828), 262–269. https://doi.org/10.1038/s41586-020-2781-z

Lesch, K.-P., Timmesfeld, N., Renner, T. J., Halperin, R., Röser, C., Nguyen, T. T., Craig, D. W., Romanos, J., Heine, M., Meyer, J., Freitag, C., Warnke, A., Romanos, M., Schäfer, H., Walitza, S., Reif, A., Stephan, D. A., & Jacob, C. (2008). Molecular genetics of adult ADHD: Converging evidence from genome-wide association and extended pedigree linkage studies. *Journal of Neural Transmission*, *115*(11), 1573–1585. https://doi.org/10.1007/s00702-008-0119-3

Levey, D. F., Gelernter, J., Polimanti, R., Zhou, H., Cheng, Z., Aslan, M., Quaden, R., Concato, J., Radhakrishnan, K., Bryois, J., Sullivan, P. F., the Million Veteran Program, & Stein, M. B. (2020). Reproducible Genetic Risk Loci for Anxiety: Results From ∼200,000 Participants in the Million Veteran Program. *American Journal of Psychiatry*, *177*(3), 223–232. https://doi.org/10.1176/appi.ajp.2019.19030256

Machado, C. J., & Bachevalier, J. (2008). Behavioral and hormonal reactivity to threat: Effects of selective amygdala, hippocampal or orbital frontal lesions in monkeys. *Psychoneuroendocrinology*, *33*(7), 926–941.

https://doi.org/10.1016/j.psyneuen.2008.04.012

Miller, C. T., Freiwald, W. A., Leopold, D. A., Mitchell, J. F., Silva, A. C., & Wang, X. (2016). Marmosets: A Neuroscientific Model of Human Social Behavior. *Neuron*, *90*(2), 219–233. https://doi.org/10.1016/j.neuron.2016.03.018

Mueller, S. A. L., Oler, J. A., Roseboom, P. H., Aggarwal, N., Kenwood, M. M., Riedel, M. K., Elam, V. R., Olsen, M. E., DiFilippo, A. H., Christian, B. T., Hu, X., Galvan, A., Boehm, M. A., Michaelides, M., & Kalin, N. H. (2023). DREADD-mediated amygdala activation is sufficient to induce anxiety-like responses in young nonhuman primates. *Current Research in Neurobiology*, *5*, 100111. https://doi.org/10.1016/j.crneur.2023.100111

Murray, E. A., & Izquierdo, A. (2007). Orbitofrontal cortex and amygdala contributions to affect and action in primates. *Annals of the New York Academy of Sciences*, *1121*, 273–296. https://doi.org/10.1196/annals.1401.021

Nagel, M., Jansen, P. R., Stringer, S., Watanabe, K., Leeuw, C. A. de, Bryois, J., Savage, J. E., Hammerschlag, A. R., Skene, N. G., Muñoz-Manchado, A. B., White, T., Tiemeier, H., Linnarsson, S., Hjerling-Leffler, J., Polderman, T. J. C., Sullivan, P. F., Sluis, S. van der, & Posthuma, D. (2018). Meta-analysis of genome-wide association studies for neuroticism in 449,484 individuals identifies novel genetic loci and pathways. *Nature Genetics*, *50*(7), 920–927. https://doi.org/10.1038/s41588-018-0151-7

Oler, J. A., Birn, R. M., Patriat, R., Fox, A. S., Shelton, S. E., Burghy, C. A., Stodola, D. E., Essex, M. J., Davidson, R. J., & Kalin, N. H. (2012). Evidence for coordinated functional activity within the extended amygdala of non-human and human primates. *NeuroImage*, *61*(4), 1059–1066. https://doi.org/10.1016/j.neuroimage.2012.03.045

Oler, J. A., Fox, A. S., Shelton, S. E., Rogers, J., Dyer, T. D., Davidson, R. J., Shelledy, W., Oakes, T. R., Blangero, J., & Kalin, N. H. (2010). Amygdalar and hippocampal substrates of anxious temperament differ in their heritability. *Nature*, *466*(7308), 864–868. https://doi.org/10.1038/nature09282

Öngür, D., & Price, J. L. (2000). The Organization of Networks within the Orbital and Medial Prefrontal Cortex of Rats, Monkeys and Humans. *Cerebral Cortex*, *10*(3), 206–219. https://doi.org/10.1093/cercor/10.3.206

Paus, T., Collins, D. L., Evans, A. C., Leonard, G., Pike, B., & Zijdenbos, A. (2001). Maturation of white matter in the human brain: A review of magnetic resonance studies. *Brain Research Bulletin*, *54*(3), 255–266.

Petrides, M., & Pandya, D. N. (1999). Dorsolateral prefrontal cortex: Comparative cytoarchitectonic analysis in the human and the macaque brain and corticocortical connection patterns: Dorsolateral prefrontal cortex in human and monkey. *European Journal of Neuroscience*, *11*(3), 1011–1036. https://doi.org/10.1046/j.1460-9568.1999.00518.x

Phan, K. L., Orlichenko, A., Boyd, E., Angstadt, M., Coccaro, E. F., Liberzon, I., & Arfanakis, K. (2009). Preliminary Evidence of White Matter Abnormality in the Uncinate Fasciculus in Generalized Social Anxiety Disorder. *Biological Psychiatry*, *66*(7), 691–694. https://doi.org/10.1016/j.biopsych.2009.02.028

Pine, D. S., Wise, S. P., & Murray, E. A. (2021). Evolution, Emotion, and Episodic Engagement. *American Journal of Psychiatry*, *178*(8), 701–714. https://doi.org/10.1176/appi.ajp.2020.20081187

Price, J. L. (2007). Definition of the Orbital Cortex in Relation to Specific Connections with Limbic and Visceral Structures and Other Cortical Regions. *Annals of the New York Academy of Sciences*, *1121*(1), 54–71. https://doi.org/10.1196/annals.1401.008

Pryluk, R., Shohat, Y., Morozov, A., Friedman, D., Taub, A. H., & Paz, R. (2020). Shared yet dissociable neural codes across eye gaze, valence and expectation. *Nature*, *586*(7827), 95–100. https://doi.org/10.1038/s41586-020-2740-8

Pujara, M. S., Rudebeck, P. H., Ciesinski, N. K., & Murray, E. A. (2019). Heightened Defensive Responses Following Subtotal Lesions of Macaque Orbitofrontal Cortex. *The Journal of*

Neuroscience, *39*(21), 4133–4141. https://doi.org/10.1523/JNEUROSCI.2812-18.2019

Qi, C., Roseboom, P. H., Nanda, S. A., Lane, J. C., Speers, J. M., & Kalin, N. H. (2010). Anxiety-related behavioral inhibition in rats: A model to examine mechanisms underlying the risk to develop stress-related psychopathology. *Genes, Brain and Behavior*, *9*(8), 974–984. https://doi.org/10.1111/j.1601-183X.2010.00636.x

Rahman, S. S., Mulvihill, K., Wood, C. M., Quah, S. K. L., Horst, N. K., Clarke, H. F., Cockcroft, G. J., Santangelo, A. M., & Roberts, A. C. (2021). Differential Contribution of Anterior and Posterior Midcingulate Subregions to Distal and Proximal Threat Reactivity in Marmosets. *Cerebral Cortex*, *31*(10), 4765–4780. https://doi.org/10.1093/cercor/bhab121

Rich, E. L., & Wallis, J. D. (2016). Decoding subjective decisions from orbitofrontal cortex. *Nature Neuroscience*, *19*(7), 973–980. https://doi.org/10.1038/nn.4320

Roberts, A. C., & Clarke, H. F. (2019). Why we need nonhuman primates to study the role of ventromedial prefrontal cortex in the regulation of threat- and reward-elicited responses. *Proceedings of the National Academy of Sciences*, *116*(52), 26297–26304. https://doi.org/10.1073/pnas.1902288116

Rogers, J., Raveendran, M., Fawcett, G. L., Fox, A. S., Shelton, S. E., Oler, J. A., Cheverud, J., Muzny, D. M., Gibbs, R. A., Davidson, R. J., & Kalin, N. H. (2013). CRHR1 genotypes, neural circuits and the diathesis for anxiety and depression. *Molecular Psychiatry*, *18*(6), 700–707. https://doi.org/10.1038/mp.2012.152

Roseboom, P. H., Mueller, S. A. L., Oler, J. A., Fox, A. S., Riedel, M. K., Elam, V. R., Olsen, M. E., Gomez, J. L., Boehm, M. A., DiFilippo, A. H., Christian, B. T., Michaelides, M., & Kalin, N. H. (2021). Evidence in primates supporting the use of chemogenetics for the treatment of human refractory neuropsychiatric disorders. *Molecular Therapy*, *29*(12), 3484–3497. https://doi.org/10.1016/j.ymthe.2021.04.021

Rudebeck, P. H., Saunders, R. C., Prescott, A. T., Chau, L. S., & Murray, E. A. (2013).

Prefrontal mechanisms of behavioral flexibility, emotion regulation and value updating. *Nature Neuroscience*, *16*(8), 1140–1145. https://doi.org/10.1038/nn.3440

Saleem, K. S., Kondo, H., & Price, J. L. (2008). Complementary circuits connecting the orbital and medial prefrontal networks with the temporal, insular, and opercular cortex in the macaque monkey. *The Journal of Comparative Neurology*, *506*(4), 659–693. https://doi.org/10.1002/cne.21577

Schmitz, M. T., Sandoval, K., Chen, C. P., Mostajo-Radji, M. A., Seeley, W. W., Nowakowski, T. J., Ye, C. J., Paredes, M. F., & Pollen, A. A. (2022). The development and evolution of inhibitory neurons in primate cerebrum. *Nature*, *603*(7903), 871–877. https://doi.org/10.1038/s41586-022-04510-w

Schwartz, C. E., Wright, C. I., Shin, L. M., Kagan, J., & Rauch, S. L. (2003). Inhibited and uninhibited infants “grown up”: Adult amygdalar response to novelty. *Science (New York, N.Y.)*, *300*(5627), 1952–1953. https://doi.org/10.1126/science.1083703

Shackman, A. J., & Fox, A. S. (2016). Contributions of the Central Extended Amygdala to Fear and Anxiety. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, *36*(31), 8050–8063. https://doi.org/10.1523/JNEUROSCI.0982-16.2016

Shackman, A. J., Fox, A. S., Oler, J. A., Shelton, S. E., Oakes, T. R., Davidson, R. J., & Kalin, N. H. (2017). Heightened extended amygdala metabolism following threat characterizes the early phenotypic risk to develop anxiety-related psychopathology. *Molecular Psychiatry*, *22*(5), 724–732. https://doi.org/10.1038/mp.2016.132

Smith, D. J., Escott-Price, V., Davies, G., Bailey, M. E. S., Colodro-Conde, L., Ward, J., Vedernikov, A., Marioni, R., Cullen, B., Lyall, D., Hagenaars, S. P., Liewald, D. C. M., Luciano, M., Gale, C. R., Ritchie, S. J., Hayward, C., Nicholl, B., Bulik-Sullivan, B., Adams, M., … O’Donovan, M. C. (2016). Genome-wide analysis of over 106 000 individuals identifies 9 neuroticism-associated loci. *Molecular Psychiatry*, *21*(6), 749–757. https://doi.org/10.1038/mp.2016.49

Stawicka, Z. M., Massoudi, R., Oikonomidis, L., McIver, L., Mulvihill, K., Quah, S. K. L., Cockcroft, G. J., Clarke, H. F., Horst, N. K., Wood, C. M., & Roberts, A. C. (2022). Differential Effects of the Inactivation of Anterior and Posterior Orbitofrontal Cortex on Affective Responses to Proximal and Distal Threat, and Reward Anticipation in the Common Marmoset. *Cerebral Cortex*, *32*(7), 1319–1336. https://doi.org/10.1093/cercor/bhab240

Stein, M. B., Levey, D. F., Cheng, Z., Wendt, F. R., Harrington, K., Pathak, G. A., Cho, K., Quaden, R., Radhakrishnan, K., Girgenti, M. J., Ho, Y.-L. A., Posner, D., Aslan, M., Duman, R. S., Zhao, H., Department of Veterans Affairs Cooperative Studies Program (no. 575B), Stein, M. B., Levey, D. F., Cheng, Z., … Gelernter, J. (2021). Genome-wide association analyses of post-traumatic stress disorder and its symptom subdomains in the Million Veteran Program. *Nature Genetics*, *53*(2), 174–184. https://doi.org/10.1038/s41588-020-00767-x

Swanson, L. W., & Petrovich, G. D. (1998). What is the amygdala? *Trends in Neurosciences*, *21*(8), 323–331.

Tovote, P., Esposito, M. S., Botta, P., Chaudun, F., Fadok, J. P., Markovic, M., Wolff, S. B. E., Ramakrishnan, C., Fenno, L., Deisseroth, K., Herry, C., Arber, S., & Lüthi, A. (2016). Midbrain circuits for defensive behaviour. *Nature*, *534*(7606), 206–212. https://doi.org/10.1038/nature17996

Tromp, D. P. M., Fox, A. S., Oler, J. A., Alexander, A. L., & Kalin, N. H. (2019). The Relationship Between the Uncinate Fasciculus and Anxious Temperament Is Evolutionarily Conserved and Sexually Dimorphic. *Biological Psychiatry*, *86*(12), 890–898. https://doi.org/10.1016/j.biopsych.2019.07.022

Tromp, D. P. M., Grupe, D. W., Oathes, D. J., McFarlin, D. R., Hernandez, P. J., Kral, T. R. A., Lee, J. E., Adams, M., Alexander, A. L., & Nitschke, J. B. (2012). Reduced Structural Connectivity of a Major Frontolimbic Pathway in Generalized Anxiety Disorder. *Archives*

of General Psychiatry, *69*(9), 925. https://doi.org/10.1001/archgenpsychiatry.2011.2178

Tromp, D. P. M., Williams, L. E., Fox, A. S., Oler, J. A., Roseboom, P. H., Rogers, G. M., Benson, B. E., Alexander, A. L., Pine, D. S., & Kalin, N. H. (2019). Altered Uncinate Fasciculus Microstructure in Childhood Anxiety Disorders in Boys But Not Girls. *The American Journal of Psychiatry*, appiajp201818040425. https://doi.org/10.1176/appi.ajp.2018.18040425

Wallis, J. D. (2012). Cross-species studies of orbitofrontal cortex and value-based decision-making. *Nature Neuroscience*, *15*(1), 13–19. https://doi.org/10.1038/nn.2956

Welker, B. J. (2017). *The history of our tribe, Hominini*. Open SUNY Textbooks.

Wellman, L. L., Forcelli, P. A., Aguilar, B. L., & Malkova, L. (2016). Bidirectional Control of Social Behavior by Activity within Basolateral and Central Amygdala of Primates. *The Journal of Neuroscience*, *36*(33), 8746–8756. https://doi.org/10.1523/JNEUROSCI.0333-16.2016